A Third Millennium Levantine Pottery Production Center: Typology, Petrography, and Provenance of the Metallic Ware of Northern Israel and Adjacent Regions

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Although long considered a hallmark of the EB II–III of northern Canaan, Metallic Ware (sometimes termed Abydos Ware, Combed Ware) has eluded systematic characterization. Defined by its highly fired fabric, Metallic Ware comprises a full range of household forms, excluding cooking pots. It is widely distributed between Ta^canakh in the south and Tyre in the north, and from Khirbet ez-Zeraqun in the east to the Mediterranean coast. Wherever found, it exhibits a unity of typology, chronology (floruit in EB II, decline in EB III), and fabric. Sherds from eight sites were analyzed petrographically, revealing a similar geological provenance: Lower Cretaceous formations that crop out mainly in the Hermon massif and north, in Lebanon. In view of its stylistic affinity to contemporary Canaanite pottery, it is proposed that Metallic Ware was produced in workshops centered around the upper Jordan Valley and distributed from there, in large quantities, to sites as far away as 100 km. Its pattern of distribution reflects a highly integrated, perhaps centralized, economy in EB II in northern Canaan and adjacent regions.

INTRODUCTION

Bronze Age ceramic production in the southern Levant is generally characterized by a pronounced regionalism in detailed typology and choice of raw materials, and a tendency toward local production in villages or, during urbanized periods, in or near major towns (Wood 1990: 70–85). Occasionally, specific classes of pottery are widely distributed from a specific center of manufacture (Amiran, Beit-Arieh, and Glass 1973; Beck 1985; Falconer 1987; Esse 1989). It would be rare to find a comprehensive pottery industry mass-producing its ware at a single locale and distributing it in large quantities over a considerable distance, to the extent that local industries are virtually obliterated within a large radius from the source of production. The Early Bronze Age Metallic Ware of the upper Jordan Valley and the adjacent regions of northern Transjordan, the Golan, the Galilee, and the Lebanese Biqa^c appears to provide an example of such a rare case. Its appearance in conjunction with the initial urbanization of northern Canaan and neighboring lands suggests that a study of this industry may provide further insight into the far-reaching changes effected at that time in the social fabric of these regions.

WARE DESCRIPTION AND TYPOLOGY

Although long considered a hallmark of the EB II–III in northern Israel and adjacent regions, Metallic Ware has yet to be systematically described in the archaeological literature. Various overlapping terms have been used to designate it, most notably Combed Ware and Abydos Ware (Prausnitz 1954; Mazzoni 1987; Esse 1991: 109–16; Ben-Tor 1991: 107–9), but its identification and definition as a specific class of ceramics has remained somewhat vague and intuitive. This may well be because, until very recently, no Early Bronze site of northern Canaan, where Metallic Ware is most prominent, has been published in detail.

The present study is based on our typological and petrographic study of several assemblages in northern Israel, including those described in the forthcoming or recent publications concerning Tel Dan, Hazor, and Bet Ha^{c}emeq (Greenberg in press a, b; Givon 1993), as well as unpublished finds from Bet Yerah, Tel Te$^\supset$o, Rosh Haniqra, Gamla, Tel Qashish, and Tel Yoqne^{c}am (described in Greenberg 1987; Porat 1989a). Numerous other collections, both from excavated sites and surface collections, have been consulted.

FIELD DEFINITION

The unifying feature of Metallic Ware—in all its variant forms and at all sites where it has been identified—is its fabric, that is, the composition and firing of the paste of which it is made. Metallic Ware sherds and vessels appear in varying shades of red, with thin vessels verging toward buff or gray. Occasionally, thicker vessels appear in gray or brown, indicating firing in reducing conditions. Metallic Ware is brittle to the touch, sounding a distinctive ring when struck. In section, it is generally uniform in appearance, rather coarse grained and evenly fired, with gray cores appearing mainly in large vessels. Dominant tempers vary in color from red to gray to black (see "shales" in the petrographic description, below). Carbonates (white grits) are somewhat less prevalent, and there is no use of organic temper. There is little decoration, save for pattern combing, thin slips, and continuous (or, rarely, patterned) matte burnish. Vessels are generally thinner walled than similar forms in non-Metallic Ware.

TYPOLOGY

The range of vessels produced in Metallic Ware answered all household and small-industrial needs, with the notable exception of open-flame cooking, for which Metallic Ware is ill-suited (the mechanical qualities of the ware are discussed in detail below). Vessels range in size from the 2 cm deep saucer to the 60 cm deep "vat," and from the diminutive juglet to the meter-high pithos (figs. 1–3). In addition to these utilitarian items, symbolic objects such as animal figurines and bed models (fig. 4) were produced in Metallic Ware. Nearly all known EB II–III seal impressions of the geometric and cultic type were impressed on Metallic Ware jars or pithoi (Ben-Tor 1978: 101; Esse 1990: 32).

The following is a typological abstract of the principal forms in Metallic Ware. It relies largely on the stratified assemblage from Tel Dan (Greenberg in press a); however, as there are lapses in the Tel Dan assemblage, the range of forms is rounded out by vessels from additional sites. Figures 1–3 show the drawings of the prototypes in typological order; captions therein identify the sites at which the vessels originate.

- B1: Saucers
  - a. Shallow (fig. 1:1–2)
  - b. Hemispheric
    1. Plain rim (fig. 1:3)
    2. Flattened rim (fig. 1:4–5)
  - c. Carinated (fig. 1:6)
- B2: Platter-bowls
  - a. Shallow (fig. 1:7–10)
  - b. Deep
    1. Smooth interior (fig. 1:11–12)
    2. Striated interior (grater? cf. London 1992) (fig. 1:13)
- J1: Jugs and juglets, rim-handle
  - a. Slender (fig. 1:14–15)
  - b. Squat (fig. 1:16)
- J2: Jugs, neck-handle (fig. 1:17)
- SmJ1: Small jar, channeled rim, lug-handles (fig. 1:18)
- SmJ2: Small jar, plain rim, no handles (fig. 1:19)
- V1: Combed vats, small (D < 30 cm) (fig. 2:1)
- V2: Combed vats, large (D > 40 cm) (fig. 2:2)
- SJ1: Combed jar (H < 60 cm)
  - a. Slender (fig. 2:3–4)
  - b. Squat (fig. 2:5–6)
- SJ2: Channeled-rim jar, pattern burnished (fig. 3:1)
- SJ3: Combed pithos (H > 80 cm)
  - a. Thickened rim (fig. 3:2–3)
  - b. Broad folded rim, decorated neck (fig. 3:4–5)

Remarks: Because Metallic Ware was handmade, there are endless variations in detailed typology, most of which cannot (indeed, should not) be addressed individually. The following attributes should, however, be noted: (a) Red slip is rather more common on bowls than would appear in fig. 1. It is most often applied on the interior and rim. (b) While continuous burnish on bowl rims is common, pattern burnish is rare, and more often found on jugs and jars than on bowls. (c) A tool-cut concavity on platter exteriors, below the rim, is quite common. (d) Vats were apparently spouted, although extant fragments do not always reflect this.

Several methods were used in the construction of Metallic Ware vessels: bowls and platters were formed in a mold or by the upside-down method, in

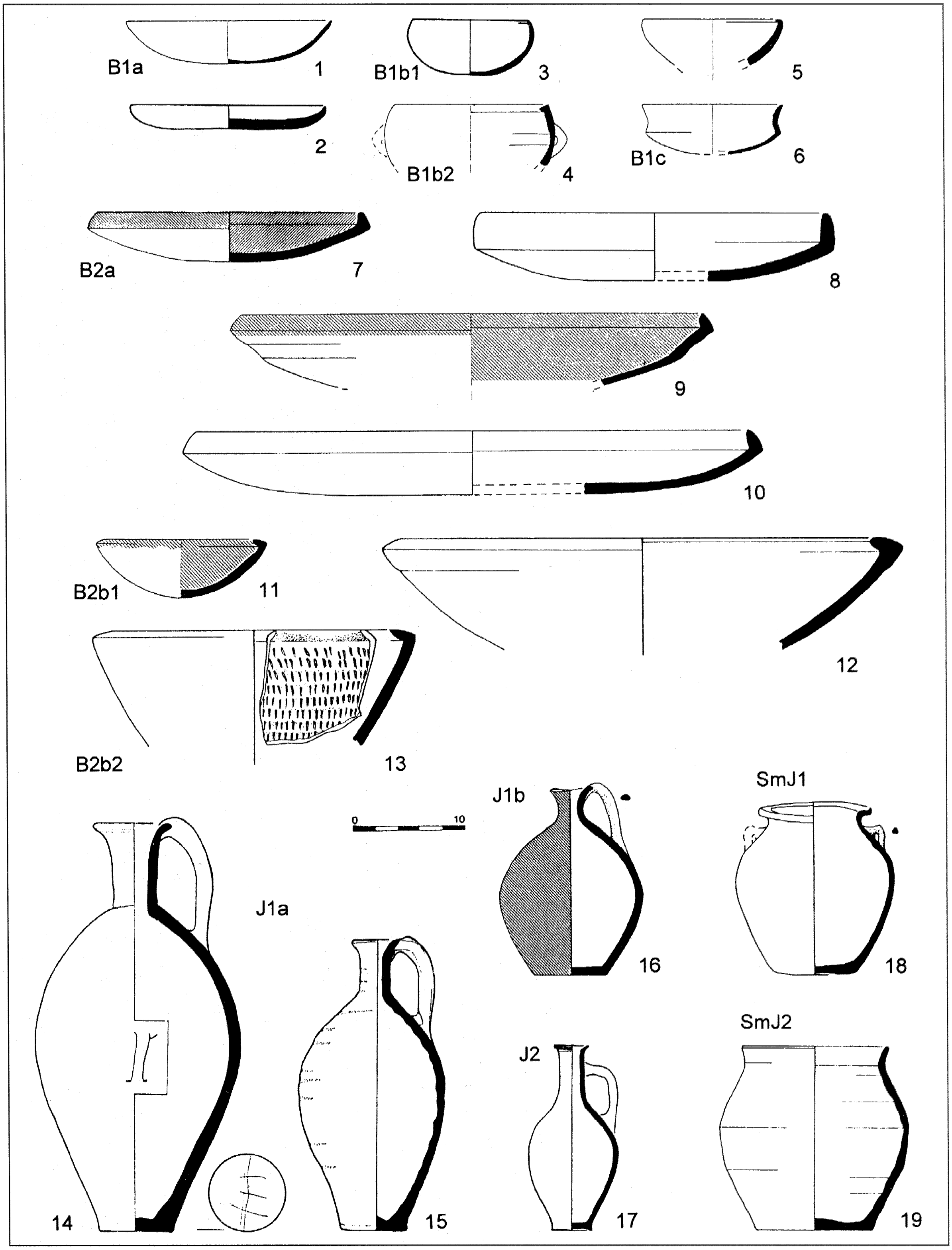

Fig. 1. Metallic Ware typology. Sources: 1, 4–6, 9, 11: Tel Teʾo, Stratum III; 2: Tel Dan, Area K, Locus 6330b; 3: Hazor, Area L, Locus 1161; 7, 10, 16, 18–19: Tel Dan, Phase B7a; 8: Tel Dan, Phase B6; 12: Tel Dan, Phase B7b; 13: Tel Dan, Phase M4; 14–15, 17: Kinneret Tomb.

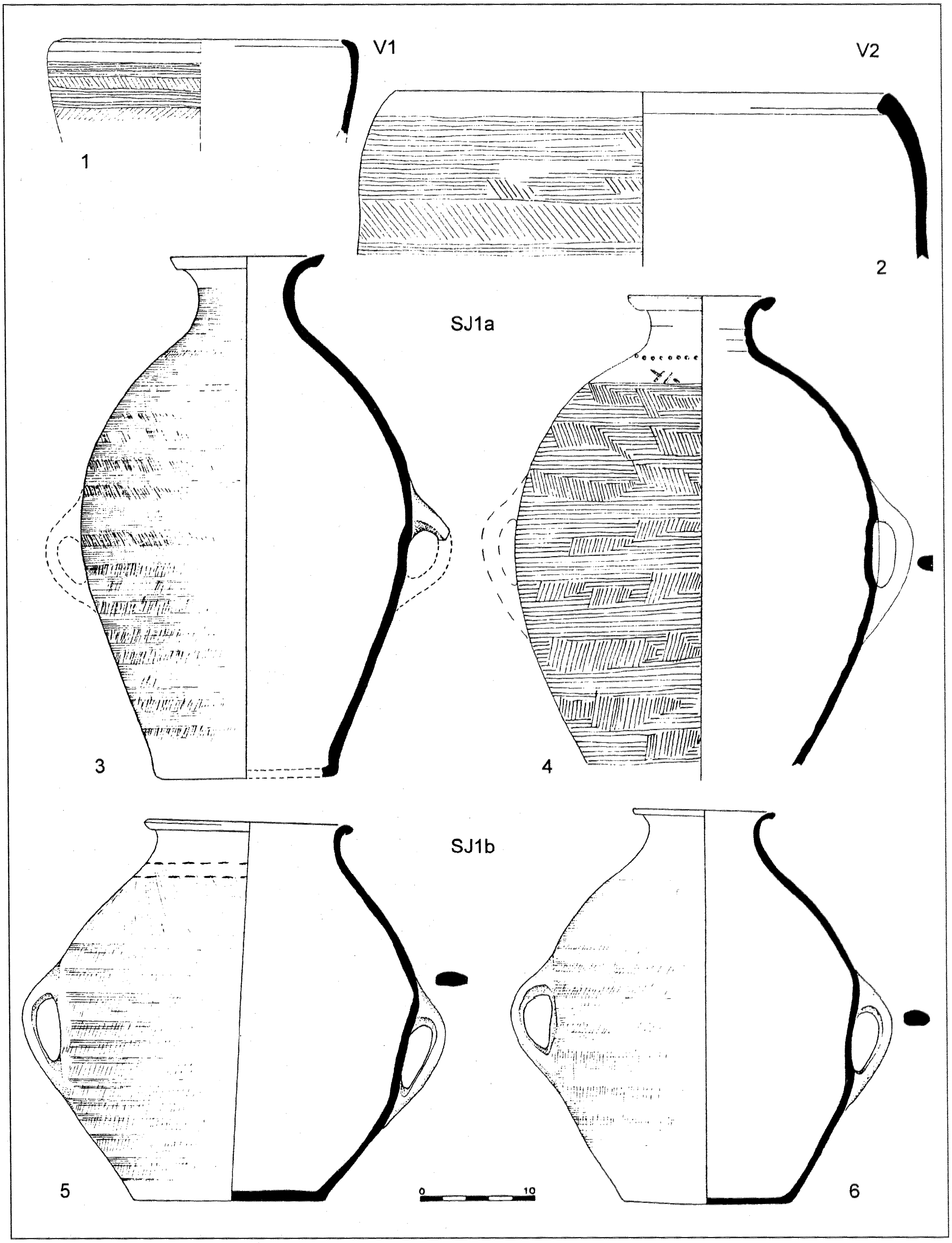

Fig. 2. Metallic Ware typology (continued). Sources: 1: Tel Teᵓo, Stratum III; 2, 6: Tel Dan, Phase B7a; 3: Tel Dan, Phase B6; 4: Tel Dan, Phase B7b; 5: Tel Dan, Phase Y2.

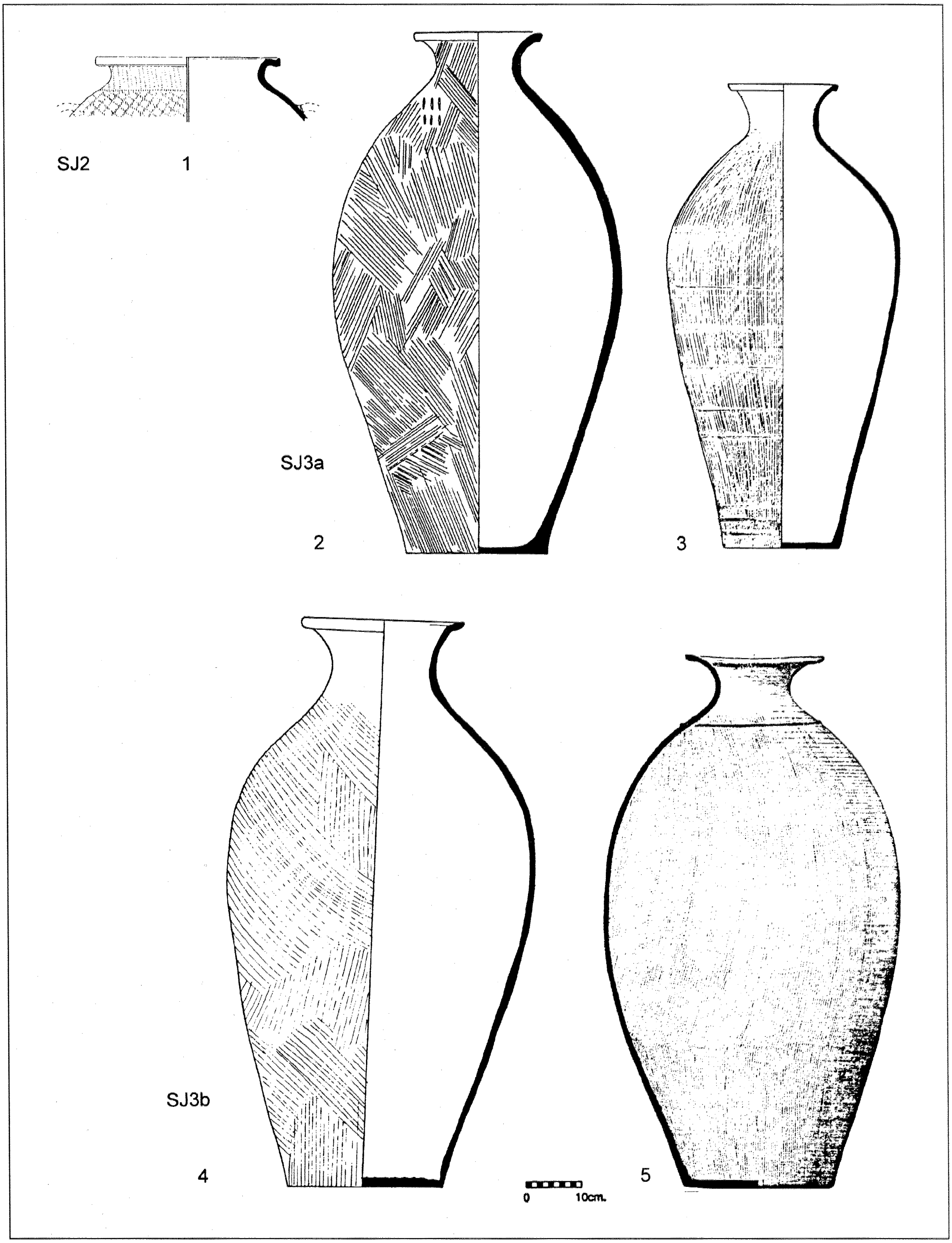

Fig. 3. Metallic Ware typology (continued). Sources: 1: Tel Dan Phase B7b; 2: Bet Ha^{c}emeq, Stratum I (Givon 1993: fig. 15:1); 3: Tel Qashish Area A, Locus 126; 4: Hazor, Area A, Locus 636; 5: Khirbet ez-Zeraqun, HZ87–323.

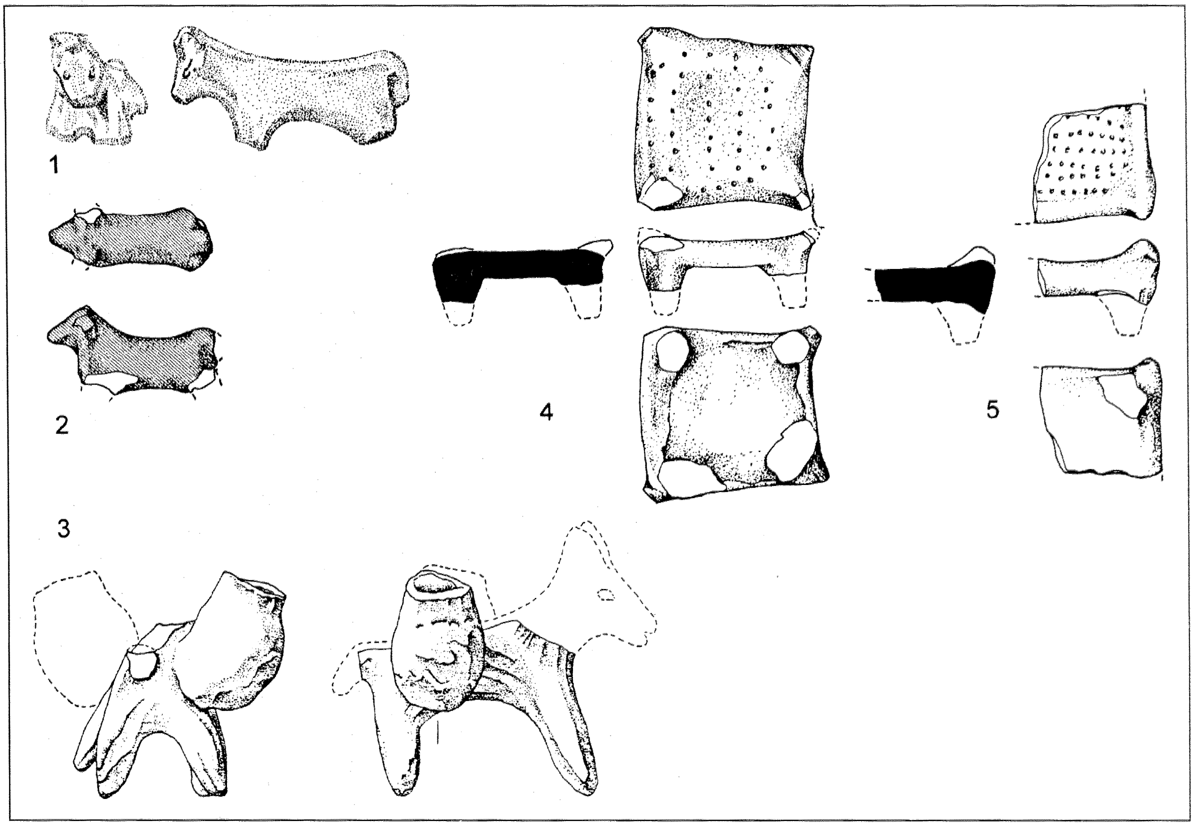

Fig. 4. Figurines and bed models in Metallic Ware from Tel Dan. The laden donkey (No. 3) could illustrate the manner in which Metallic Ware pottery was transported. Scale approximately 1:2.

which bowl interiors are formed on a thick slab of clay which is subsequently turned over and pared down to size. Jugs and larger containers were built in coils, from the base up (cf. the description of similar manufacturing techniques at EB II–III Tel Yarmuth; London 1988). Vertical burnish was used to smooth and mask the joins between the coils on smaller vessels (occasionally on jars); pattern-combing served a similar purpose on jars, vats, and pithoi (the lime wash typical of combed vessels in southern Israel is not attested in Metallic Ware). Nearly all vessels (or at least their rims) were finished with a rotary motion; this is most evident on the rims of platters and jars, as well as on some jug rims and bodies. The join between jar necks and shoulders is sometimes marked by incised or impressed decoration or by applied rope decoration, particularly on pithoi.

The high firing temperatures to which Metallic Ware vessels were subjected (below) precluded the attainment of the glossy sheen so characteristic of other Early Bronze wares (cf. London 1988: 122). The resulting matte burnish was therefore only rarely applied in a pattern, although a handsome effect occasionally was produced in this manner (fig. 3:1). Potter's marks are common, particularly on the bases of platters and jugs, and on the shoulders of jars (figs. 1:14, 2:4, 3:2).

In terms of its ceramic tradition (typology plus technique) Metallic Ware duplicates, almost in its entirety, the Early Bronze repertoire of non-Metallic ceramic assemblages (see, typically, Amiran 1969: pls. 15–16), with a few innovations (Type B1a saucers; Type V1, 2 vats). However, the corpus as a whole suggests a more determined effort at streamlined symmetry and spare functionalism. The vessels were plainly made with an eye toward the needs of transport: they are relatively light, due to their thin wall; even the largest pithos can be easily packed (fig. 4:3); and there are no outsized platters (contrast, e.g., de Miroschedji 1988: pl. 43). Among the jars, holemouth jars, which are difficult to seal, are avoided, as are ledge handles; these are replaced by the far more durable loop handles, streamlined by virtue of the characteristic concavity in the jar wall opposite the handle.

DISTRIBUTION

Metallic Ware is abundant in all EB II–III sites surveyed or excavated in the following regions: the Hula and upper Jordan Valleys, the southern Lebanese Biqa^c, the Golan plateau, the highlands and coast of southern Lebanon, and the Galilee and the Jezreel Valley. Figure 5 presents a digest of sites at which significant amounts of Metallic Ware have been reported.

The southernmost site reporting large quantities of Metallic Ware is Tel Ta^canakh; the easternmost is Khirbet ez-Zeraqun in Jordan. Apart from Tel Ta^canakh, no site south of the Jezreel Valley has produced anything more than isolated vessels in Metallic Ware. Isolated Metallic Ware sherds were identified (fig. 6), either by their appearance or by their petrographic composition, at Tel Dalit (Y. Goren, personal communication), Tel el-Far^cah North (de Vaux and Stève 1948: fig. 4), Ai, Tel ^cErani, Arad, and Northern Sinai (Porat 1989a: fig. 10:9; Amiran 1978: 114–15; pl. 23:7); presumably they exist at other sites, where they were not always identified as such.

As for the northern distribution of Metallic Ware, a distinction should be drawn between the Canaanite Metallic Ware assemblage and metallic vessels appearing from Sidon north, as far as Ras Shamra and Sukas. The latter, while they share many traits with Canaanite Metallic Ware, appear as components of a different assemblage with distinctive non-Canaanite traits (e.g., Saghieh 1983: pls. 35–39; Ehrich 1939: 27–29, 50, 87 [Types XI–XII], pl. 8; Oldenburg 1991; Courtois 1962; de Contenson 1969; 1979; 1992), and are made in a different fabric (below), though of similar quality (the similarity to Ehrich's description of her Stone Ware is particularly striking).

In terms of relative quantity, Metallic Ware constitutes 50 percent or more of the ceramic corpus, excluding cooking pots, in certain stratigraphic phases at the southernmost sites where it is abundant; this estimate is based on preliminary sherd counts from Tel Qashish (S. Zuckerman, personal communication), and our own observations of the excavated material from sites such as Qiryat Ata, Qishyon, and Bet Yerah. The relative quantity increases as one moves north, so that at sites such as Tel Te^ɔo and Tel Dan, Metallic Ware constitutes more than 85 percent of the chronologically relevant assemblages (based on sherd counts), the same apparently being true of sites such as Rosh Haniqra and Me^cona.

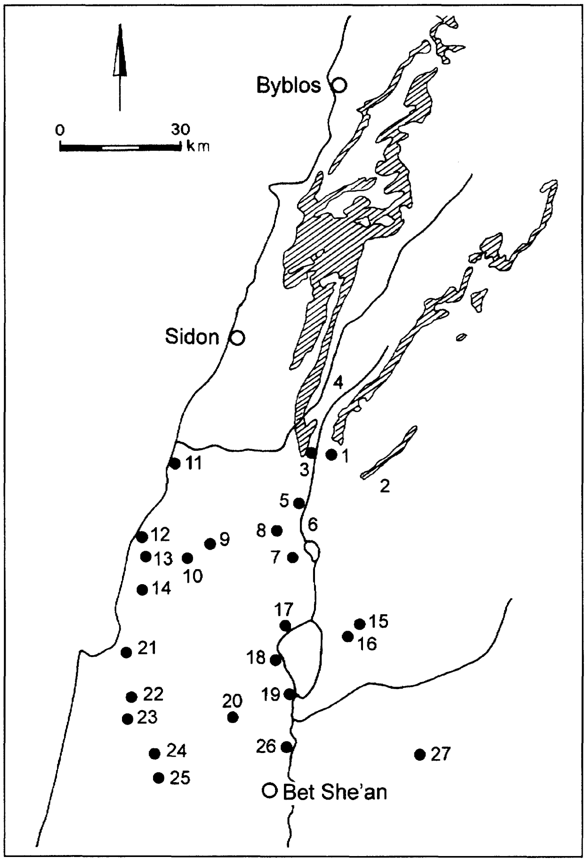

Fig. 5. Principal sites with Metallic Ware. Shaded areas indicate outcrops of the Lower Cretaceous Hatira formation. 1. Tel Dan (Greenberg in press a); 2. Northern Golan Survey (Hartal 1989: 4–5); 3. Tel Abel Bet-Ma^cacah (Dever 1986); 4. Lebanese Biqa^c Survey (Marfoe 1978; 1979); 5. Tel Te^ɔo (Greenberg 1987: 134–42); 6. Hula Valley Survey (Greenberg 1990); 7. Hazor (Greenberg in press b); 8. Tel Qedesh (Aharoni 1993); 9. Tel Rosh (er-Ruweise) (Amiran 1953); 10. Me^cona (Braun 1993); 11. Tyre (Bikai 1978: 69–70, pls. 57A, 58A); 12. Rosh Haniqra (Tadmor and Prausnitz 1959); 13. Kabri (Scheftelowitz 1990: X); 14. Bet Ha^cemeq (Givon 1993: figs. 11–13, 15); 15. Gamla (Porat 1989a: Fig. 10.9; S. Guttmann personal communication); 16. Leviah enclosure (Kochavi 1993); 17. Tel Kinnerot (Fritz 1990; 1993); 18. Tel Reqet (Khirbet el-Quneitra) (Prausnitz 1957); 19. Tel Bet Yerah (Esse 1982; 1991); 20. Tel Qishyon (Arnon and Amiran 1993); 21. Qiryat Ata (Golani and Braun 1991); 22. Tel Qashish (Ben-Tor 1993b); 23. Tel Yoqne^ɔam (Porat 1989a; Ben-Tor 1993a: 811); 24. Megiddo (Engberg and Shipton 1934: 10–11); 25. Ta^canakh (Esse 1982: 213–17); 26. Tel Yaqush (Esse 1993); 27. Khirbet ez-Zeraqun (Ibrahim and Mittmann 1987; H. Genz personal communication).

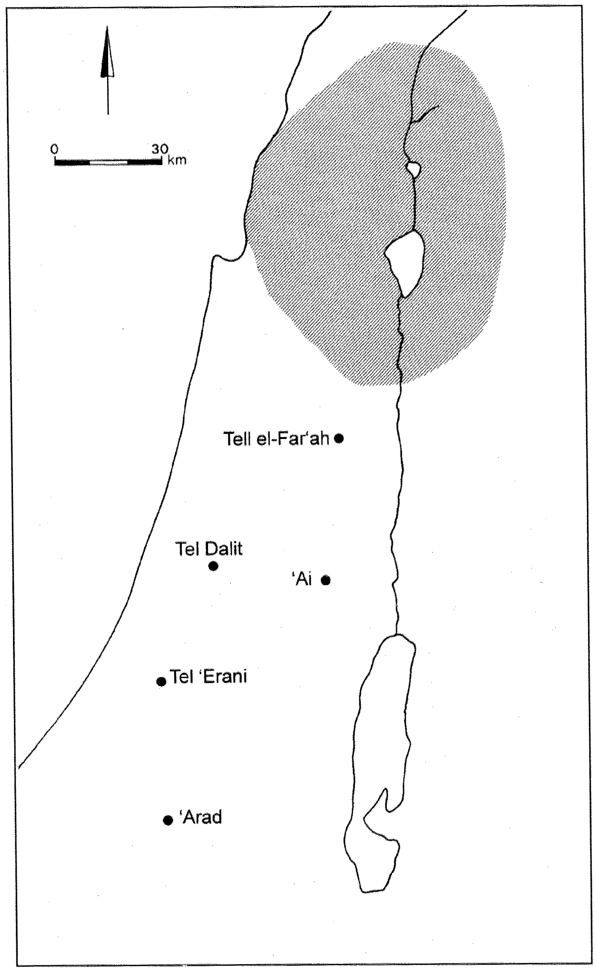

Fig. 6. Additional sites with Metallic Ware. Area with intensive distribution (see fig. 3) is shaded.

At all the excavated sites where it is abundant, Metallic Ware appears in its full range of forms, open and closed, large and small.

CHRONOLOGY: APPEARANCE AND FLORUIT OF METALLIC WARE

Secure dating for Metallic Ware is provided by isolated finds in the EB II strata at Arad and Tel el-Far^{c}ah North (Amiran 1978: 114–15; pl. 23:7; de Vaux and Stève 1948: fig. 4). A sole appearance of a Metallic Ware jar in an EB I context has been reported at Tel Yaqush (Esse 1993), but a fuller understanding of the significance of this find requires more detailed documentation. In any case, the full typological range of Metallic Ware is clearly in the EB II tradition, and has little in common with EB I forms.

At northern sites, the massive appearance of Metallic Ware is sandwiched between EB Ib phases on the one hand, and phases with large amounts of Khirbet Kerak Ware on the other. Thus, at Qiryat Ata, Bet Ha^{c}emeq, Kabri, Rosh Haniqra, Tel Qashish, Tel Qishyon, Tel Bet Yerah, Tel Te$^{\supset}$o, and other sites (see fig. 5 for references), Metallic Ware first appears, in clear stratigraphic contexts, above strata dated to the latter part of EB I. At sites lacking EB I remains, Metallic Ware appears in quantity in the lowest EB strata (e.g., Hazor and Tel Dan). At the other end of the spectrum, many sites characterized by Metallic Ware do not show evidence of EB III occupation (e.g., Qiryat Ata, Me^{c}ona, Tel Te$^{\supset}$o). At sites with EB III occupation, the relative quantity of Metallic Ware is drastically reduced, most notably the strata rich in Khirbet Kerak Ware at Qishyon, Tel Yaqush, Hazor, and Tel Bet Yerah. However, at both Hazor and Tel Qishyon, phases have been reportedly identified with isolated Khirbet Kerak sherds appearing alongside abundant Metallic Ware, suggesting a possible overlap between the two phenomena (the history of import and production of Khirbet Kerak Ware in Canaan has yet to be studied in detail).

THE END OF METALLIC WARE

While the floruit of Metallic Ware is clearly limited to EB II (and possibly early EB III), there is a growing body of evidence suggesting a limited presence of Metallic Ware throughout EB III. A complete pithos comes from an EB III phase at Hazor (fig. 3:4), Metallic Ware storejars have been found alongside Khirbet Kerak Ware at Leviah Enclosure (L. Vinitzky, personal communication) and Khirbet ez-Zeraqun (H. Genz, personal communication). At other EB III sites, combed Metallic Ware sherds are common, though their derivation, at least in part, from EB II strata at these sites is not unlikely. The sole typological development found in Metallic Ware from EB III contexts is the appearance of the sharply everted feathered-edge folded rim and rope-decorated neck on pithoi (fig. 3:5). It thus seems that Metallic Ware jars, and especially pithoi, continued to be produced in EB III, after the production of the wide range of Metallic Ware forms had ceased. Pithos production ended before the end of EB III, and the Metallic Ware tradition is not included among the ceramic carryovers into the Intermediate Bronze period.

In a parallel development, the Metallic Ware technique appears to have spread, in EB III, to southern Canaan. Many southern sites, e.g., Tel Yarmuth, Tell el-Hesi, and Lachish (Ben-Tor 1975; Fargo 1979; Tufnell 1958: pl. 62), yielded large quantities of combed, highly fired jars and pithoi, which are petrographically distinct from the northern vessels (below). However, southern Early Bronze sites with a strong local ceramic tradition beginning in EB I, such as Jericho and ᶜAi, do not adopt the Metallic Ware style.

PETROGRAPHY

The archaeological observations presented above appear to indicate the existence of a dominant ceramic industry in northern Canaan and the adjacent regions. Our petrographic study aims to provide both an independent corroboration of the unity of the phenomenon, observed at sites 100 km or more from each other, and an indication of the geographic locus of production.

The purposes of the petrographic analyses were to identify the raw materials used in pottery production, to describe their variability, and to determine the geological sources of the raw materials and the possible geographic area in which the vessels were manufactured.

Sampling and Method

A sample of 146 sherds was analyzed for this study, selected from eight northern sites at which Metallic Ware constitutes a substantial part of the pottery assemblage. Although the sample is not statistically valid, it does represent all types of vessels found at each site. Table 1 gives the number of samples, both metallic and nonmetallic, analyzed from each site.

The samples were thin-sectioned and examined under a polarizing microscope to identify the minerals constituting the temper and the silty fraction in the matrix. The volume of various components in the sherds was visually estimated using charts (Bullock 1985). One sample was analyzed using X-ray diffraction (XRD), to determine the bulk mineralogical composition and to estimate the firing temperature.

The thin sections of metallic and nonmetallic wares were compared to thin sections of raw materials collected from various localities in northern Israel. They were also compared to a large collection of thin sections of Early Bronze pottery from sites all over Israel. Identification of the source of the raw materials was based on data found in geological maps and reports from northern Israel concerning the exposed rocks and minerals in the area.

Results

Temper. Several components are found in the temper. They will be described in decreasing order of relative frequency. Shale fragments, quartz, and carbonates appear in all samples, while the remaining constituents are less common:

• **Shale fragments:** These are minute pieces derived from the shales used as raw material for the clay mixture (Whitbread 1986). The fragments were either added intentionally to the clay or were parts of the shales and were not well mixed with the remaining raw materials.

The fragments are oval to rectangular and are poorly sorted (fig. 7:1). Often they have a preferred orientation of the clay minerals and are birefringent. They comprise up to 15 percent of the volume of the sherd and reach a size of up to 2 mm. They are rich in clay minerals and are often similar to the matrix surrounding them. Their color varies from red and yellow to gray or black. Black shales are rich in minute iron oxides, finely dispersed through the clay, while red and yellow indicate the natural state of oxidation of iron within the clay. In most samples red, yellow, and black fragments occur together. The gray shale fragments result from a reducing atmosphere in the kiln and appear only in samples fired in such conditions.

Like the color, the silty components of the shale fragments also vary: some fragments lack any silt-sized components, some contain silty quartz while others are rich in silty iron oxides, often rhomboid and sometimes zoned.

• **Quartz:** Almost all samples contain quartz (fig. 7:2), either in the coarse (sand-sized) or fine (silt-sized) fraction. Many samples contain small quantities (up to 5 percent) of coarse quartz with a larger volume of fine quartz (10–15 percent). The coarse quartz is rounded to subrounded, sometimes with growth rims and coatings of iron oxides. Feldspars were not found. Tourmaline and zircon rarely occur; and no other, less stable, heavy minerals were identified, suggesting that the source of the quartz is in mature sandstones. The finer quartz is angular and poorly sorted, and its origin is the siltstone (below).

Table 1. Pottery Types and Number of Vessels Sampled from Each Site

Site	Bowls	Platters	Jars, pithoi	Jugs ("Abydos")	Juglets	Holemouth jars	Cooking pots	Others	Total
Tel Dan	**5** 1	**10** 1	**10** 1	**6** 3	**1**	1	6		**32** 13
Tel Te^ɔo	**1**	**3**	**1**	**4** 1					**9** 1
Tel Qashish		**5** 2	**4** 1	3	**1**	2	3		**13** 8
Tel Yoqne^ɔam		**1** 1	**2** 2			2	1		**3** 6
Hazor		**2**		**2** 1	1				**4** 2
Gamla		**2**	**4** 1	**1** 3		3		**1** 1	**8** 8
Rosh Haniqra		**3** 2	**3** 2	**1** 1		5			**7** 10
Bet Yerah		**1**	**5** 1	**1** 5	1	4	3	1	**7** 15
Total	**6** 1	**27** 6	**29** 8	**18** 14	**2** 2	17	13	**1** 2	**83** 63

Notes: In bold (top of each cell) is the number of Metallic Ware samples identified at that site. Below it in each square is the number of the nonmetallic samples. Note that there are no metallic holemouth jars or cooking pots.

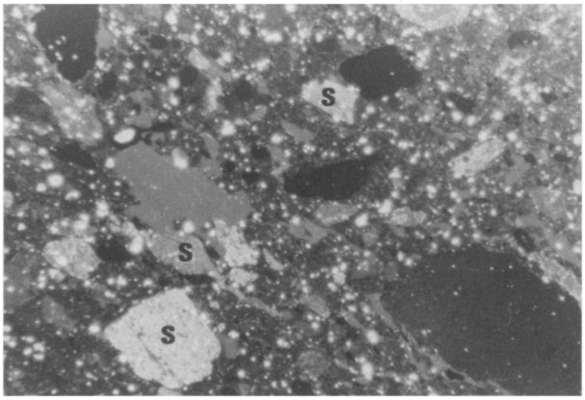

Fig. 7:1. Shale fragments (S) of various sizes and shades; the small white grains are silty quartz; a red-slipped jug from Tel Dan; W=5 mm.

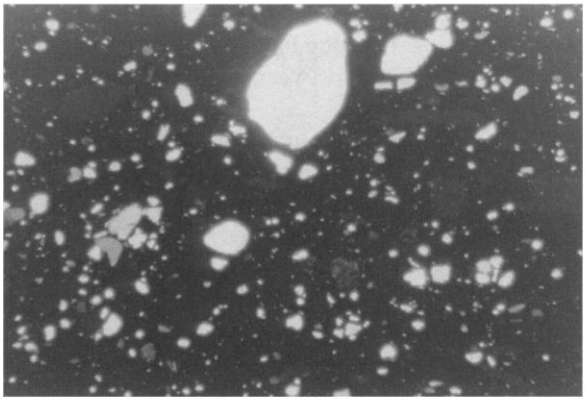

Fig. 7:2. Quartz grains; a few coarse, rounded, sand-sized grains and many smaller silt-sized grains; a large, combed jar from Tel Teᵓo; W=5 mm.

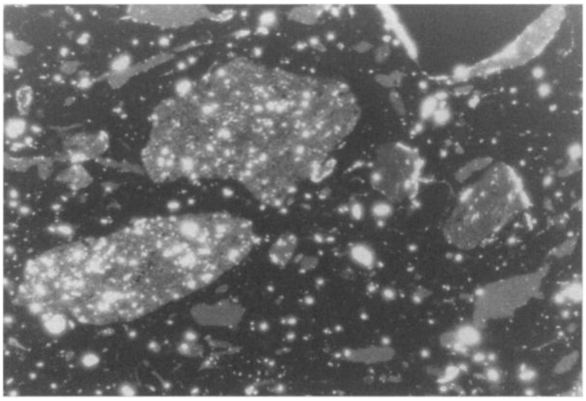

Fig. 7:3. Shale fragments rich in quartz, intermediate between siltstone and shales; a handle of a small, fine juglet from Tel Dan; W=2.5 mm.

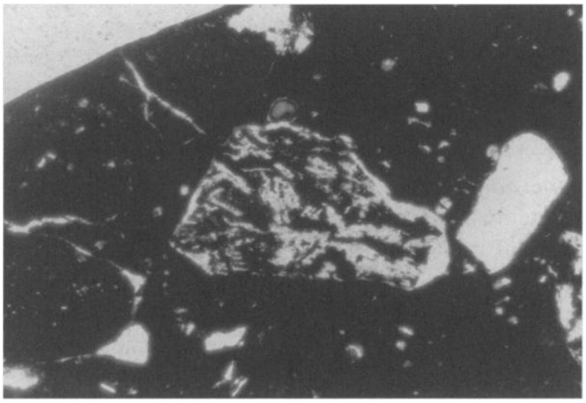

Fig. 7:4. A weathered volcanic fragment with trachytic texture; the dark phase is iron oxides and the light phase is amorphous kaolinite; at center right is a clear quartz grain and at bottom left another volcanic fragment; a deep bowl from Tel Teᵓo; W=2.5 mm.

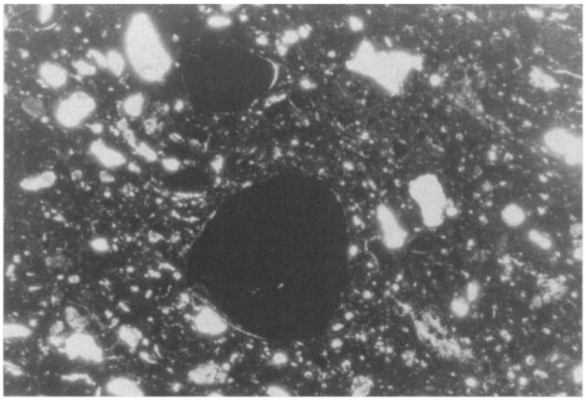

Fig. 7:5. Oolites, rounded grains of iron oxides, unique to the Lower Cretaceous Tamun formation; a burnished jar from Tel Dan; W=2.5 mm.

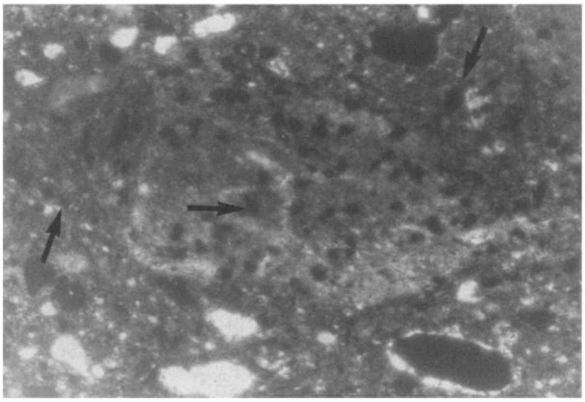

Fig. 7:6. Shale fragments and matrix rich in silt-sized iron oxides; note rhomboid shape of iron oxides (marked with arrows); a red-slipped platter from Tel Teᵓo; W=1 mm.

Fig. 7. Photographs of thin sections of Metallic Ware samples. W represents the width of the field viewed in each photograph.

Occasionally the coarse quartz grades into the fine quartz, forming one population; and in such cases there is no distinction between the temper and matrix quartz.

- **Carbonates:** Most samples contain some limestone grains, usually forming less than 5 percent of the volume of the sherd. The limestone often appears milky because it has decomposed to varying degrees, due to firing temperatures that exceeded 850°C. Under such conditions, the original texture of the limestone is no longer preserved, although occasionally traces of biogenic texture are observed. Some carbonate grains contain silt-sized iron oxides.
- **Siltstone:** These are rock fragments, aggregates of silty quartz cemented with clay minerals or iron oxides. Fine mica laths are occasionally found. With increasing clay mineral contents siltstone grades into shale fragments (fig. 7:3); and with increasing iron oxide content it grades into iron oxide grains.
- **Basalts and Other Volcanic Fragments:** Basalt fragments with a variety of textures are found in some samples. Some fragments have a trachytic texture, with elongated and oriented plagioclase laths. Other fragments consist of two phases, one colorless and almost isotropic, the other opaque (fig. 7:4). The textural relations of the two phases resemble that of volcanic rocks and the isotropic phase was interpreted as a noncrystalline component. The large proportion of the opaque phase suggests a mafic source. Under a reflected light microscope the colorless phase was identified as a silicate, while the opaque phase is composed of hematite and magnetite. All these textures are quite unlike anything known from the Tertiary basalts cropping out in the Galilee and Golan (Mor 1973).

Electron microprobe analyses (Porat 1989a: 68–69) showed that these fragments are highly weathered volcanic rocks, where kaolinite has replaced the plagioclase laths while the iron-bearing minerals were weathered to iron oxides.

- **Oolites:** These are round grains, usually made of iron oxides (fig. 7:5). Sometimes a concentric internal structure and a carbonate composition may be observed.

Matrix. Considerable variety is observed within the matrix. Usually it is clay-rich, showing uniform optical orientation. Occasionally it is vitrified due to a high firing temperature. The silty components are as follows:

- **Quartz:** As has been mentioned in the section on temper (above), most samples contain varying amounts of poorly sorted, angular, silty quartz (fig. 7:2), whose source is desaggregated siltstone.
- **Iron oxides:** Some samples contain substantial amounts of silty iron oxides with distinct features; they are rhomboid, often zoned, and some have a hollow center. Similar features were observed in the silty iron oxides within the shale fragments (fig. 7:6).
- **Carbonates:** In most samples containing a carbonate component it is not uniformly distributed in the clay, but rather appears as discrete, silt-sized particles. Although the pottery was fired to a high temperature, the reaction between the carbonate particles and the clay was not completed and the grains are still visible.
- **Mica laths:** A small number of samples contain minute muscovite mica laths, resembling those found in the siltstone. Their source is probably in the siltstone.

XRD analyses. Anorthite and gehlenite were identified in the sample analyzed by X-ray diffraction. These minerals are not found naturally in rocks and were formed within the sherd during firing by reactions between the clay minerals and the carbonate component.

Firing temperature. The firing temperature of Metallic Ware was higher than the norm for Early Bronze pottery (Porat 1989a: 62). Anorthite and gehlenite indicate a firing temperature higher than 950°C (Maggetti 1981; Maggetti, Westley, and Olin 1984). The decomposed carbonate and the vitrified clay matrix attest to temperatures higher than 850°C. The firing was mostly oxidizing, although some vessels were fired in a reducing atmosphere.

Sources of Raw Materials

The geological source of the raw materials used for producing the Metallic Ware is a siliclastic formation, composed of noncalcareous clays, siltstones, and mature sandstone, all rich in iron oxides and associated with highly weathered basalts. The formations that best suit this description are the basal Lower Cretaceous formations, which overlie Lower Cretaceous flows of weathered basalts (Tayasir Volcanics [Mimran 1969; Shaliv 1972; Bonen 1980; Shimron 1989]). While these formations crop out in small areas in the Negev, Samaria, east of the Dead Sea, and the eastern slopes of the Galilee hills, the large outcrops are found in the Hermon massif and north, in Lebanon (fig. 5).

Each of the various components found in Metallic Ware can be assigned to one of the Lower Cretaceous formations: quartz, siltstone, shale fragments, and noncalcareous, silty clays originate in the Hatira ("Nubian Sandstone") formation. The weathered basalt fragments are derived from the Tayasir Volcanics. Distinctive rhomboid-shaped silty iron oxides ("limonitic dedolomite") are found in the Faria Formation; and biogenic limestone, calcareous clays, and oolites are described from the Natufa and Tamun Formations (Mimran 1969: 27–29, fig. 25; Rohrlich 1972), above the Hatira Formation. Soils developed on Lower Cretaceous sandy bedrock contain iron-oxide-coated quartz (Singer and Amiel 1974) like that observed in the Metallic Ware.

The most common basalts in Israel, covering extensive areas of the Golan and Galilee, are Cenozoic. These differ petrographically from the basalts seen in the Metallic Ware thin sections and they do not constitute the raw materials used for Metallic Ware. Also, these Cenozoic basalts are not associated with mature sandstone, siltstone, and noncalcareous clays.

Qualities of the Raw Material

Lower Cretaceous clays were favored for pottery production as early as the Ceramic Neolithic (e.g., Munhata [Goren 1992]; Tel Tzaf [Goren 1991: 113–14; Gophna and Sadeh 1988–1989]). In EB I, small outcrops in Wadi Far^{c}ah were utilized for producing Gray-Burnished bowls (Goren 1990). This formation also was used abundantly in later periods. Petrographic analyses of Hellenistic and Roman period pottery from the northern Golan (Porat 1989b) have shown that the locally produced Ituraean pottery was manufactured from raw materials similar to those used for Metallic Ware. The Ituraean pottery contains quartz, siltstone, shale fragments, and iron oxides, and evidently it was made from the locally exposed Lower Cretaceous Hatira Formation.

The advantage of this formation for ceramic production lies in the composition of its clay minerals and the presence of iron. The dominant clay mineral is kaolinite (which in its pure form is used for the production of porcelain). This clay does not swell and shrink upon wetting and drying and can be fired to high temperatures to form an impermeable body. Kaolinite-rich clays are rare in Israel: in the more common clays other minerals predominate and kaolinite is only a minor component. Apparently, its superiority with respect to shrinkage was recognized in early times and it was used wherever available. The more advanced EB II industry further exploited the qualities of this clay, firing it to high temperatures to form impermeable, thin-walled, durable vessels. Iron acts as a flux, lowering the temperature at which the clay sinters into a hard body.

These qualities are not, however, suitable for cooking pots. Here other requirements must be met, particularly resistance to rapid heating and cooling. For cooking pots, other raw materials, similar to those used elsewhere in Canaan in the same period, were considered more suitable (Porat 1989a: 45–47). The temper for cooking pots was chosen such as to have the same thermal expansion coefficient as the clay, and usually was comprised of calcite or other carbonate fragments (Porat 1989a: fig. 8.3).

Comparative Petrography

The raw materials used for Metallic Ware differ significantly from those used in contemporary nonmetallic wares. They also differ from raw materials used for pottery from EB I sites in the region (Goren 1991: 130–32). At most sites EB I and nonmetallic pottery, which consists for the most part of low- or medium-fired cooking and storage vessels, can be related to locally available raw materials, whereas Metallic Ware is derived from materials found only at a considerable distance. The outcrops of the Hatira formation closest to sites with abundant Metallic Ware occur in Mount Hermon, about 10 km west of Tel Dan, and above Qiryat Shemona, near Tel Te$^{\supset}$o and other western Hula Valley sites (see fig. 5).

A further distinction may be made between northern Metallic Ware and the lime-coated combed jars, often labeled "metallic ware," which characterize southern EB III sites such as Tel Halif, Tell el-Hesi, and Tel Erani. Though fired to high temperatures, the latter are made of local calcareous, silty clays, with limestone and chalk temper (Porat 1989a: 73).

As for the north, very few petrographic studies on pottery from Lebanon and Syria are available for comparison. None of the ceramic fabrics from the north Syrian coast described by Wyckoff (1939) are similar to that of the Metallic Ware. In the Amuq basin, the predominant local paste used in all the archaeological phases contains serpentinite and actinolite, derived from metamorphic rocks (Matson in Braidwood and Braidwood 1960: 406). One class of pottery of the Amuq, the combed, metallic Brittle Orange Ware from phases G–J, shows both typological and petrographic affinity to the Metallic Ware of northern Canaan. This group contains shale fragments in a clay with silty quartz (Matson in Braidwood and Braidwood 1960: 370), not unlike the Canaanite ware. However, our analysis of a number of samples from this group has revealed, alongside the shale fragments, the presence of phyllite, schist, and serpentine, all components of metamorphic rocks, which do not appear in Metallic Ware in Israel (Porat 1989a: 73). Thus the Brittle Orange Ware appears to be but a variant of the typical local Amuq ware. The roughly contemporaneous highly fired Stone Ware of northern Syria and Mesopotamia (sometimes termed "metallic ware"; see, e.g., Schneider 1989) is completely unrelated to our Metallic Ware.

EB II carinated bowls from Afeq, characterized by Beck (1985) as metallic, have a different petrographic composition. They are made of a clay rich in silty quartz and carbonate fragments. This clay is probably related to loess and loessy soils. The metallic quality was acquired by virtue of a high firing temperature.

The question of Metallic Ware in Egypt extends beyond the purview of this study. As a preliminary remark, however, we would like to second Kantor's succinct and insightful categorization of Palestinian imports in late First Dynasty contexts (1992: 19). Her Metallic Ware (Kantor 1992: figs. 8, 14:17–18) is so like that of northern Canaan, both in shape and fabric, that the relation of the two groups can hardly be doubted. In contrast, the metallic combed ware attributed to the Old Kingdom (Kantor 1992: fig. 14: 1–16) is much closer to the Syro-Lebanese coastal varieties of Metallic Ware, typologically distinct from that discussed here.

Summary

The petrographic analysis thus indicates the following: Although there are large variations in the detailed petrographic composition of Metallic Ware vessels, the consistent appearance of most components indicates a single petrographic group. The most ubiquitous components are shale fragments and quartz. Iron oxides in various forms are also found in most samples. Taken together with the suggested source of raw materials, one can see that all the vessels were made from a single association of materials, which appear in different proportions within each vessel.

The composition of Metallic Ware and the difference between its composition and that of the non-metallic, local ware indicate that the metallic vessels found in northern Canaan were not manufactured in the immediate vicinity of most sites at which they were found. The production center must have been located at a distance, in the vicinity of the Hermon massif or perhaps further north, in Lebanon, where extensive outcrops of Hatira formation are found.

DISCUSSION

Provenance

The data presented thus far reveal three underlying uniformities of the Metallic Ware assemblage: uniformity of typology and technique, chronological uniformity, and uniformity of fabric and geological provenance. These uniformities permit us to characterize Metallic Ware production as a cohesive industry with a clear geographical focus of production.

Where was Metallic Ware manufactured? In terms of distribution, we should seek the production center nearest those sites where it is most abundant and where it appears in the greatest variety of forms. It should also be located in reasonable proximity to main trade routes and population centers, and to sources of raw material. Lastly, we should not stray too far north, in view of the close typological relationship of Metallic Ware to contemporary Canaanite forms.

All this having been said, the most likely location of the Metallic Ware ceramic industry is in the vicinity of the northern Jordan Valley sites, especially Tel Dan, the largest site north of Bet Yerah and the metropolis of the northern Jordan Valley (Greenberg 1990; Greenberg in press a; Biran 1994: 33–45). More specifically, workshops may have been located on the eastern slopes of the Naphtali hills of the Upper Galilee, or on the southern slopes of Mount Hermon.

From an ecological perspective, the Hermon foothills offer a combination of "limiting" and "permissive" factors, to cite Rice's terminology (1984: 49), which makes it an "effective ceramic environment," particularly conducive to craft specialization (at the expense of agricultural pursuits). The former ("limiting") factors include a general scarcity of arable land in a rugged topographical setting; and the latter ("permissive") factors include abundant fuel sources (wood), water, and exposed clay outcrops. Several pottery-rich EB II sites have been surveyed there by Hartal (1989), as well as a Late Roman ceramic workshop that utilized the same clay sources presumably used by Metallic Ware potters three millennia earlier (Porat 1989b). The suggested Metallic Ware workshops would have been sufficiently distant from main population centers to avoid pollution of their air and water, but near enough—within two hours' walk or so—for ready access to urban markets and traders.

Technology and Distribution

Technologically, Metallic Ware represents a quantum leap in relation to previous ceramic industries. Advanced kiln technology (and a good supply of wood for fuel) allowed the potter to maintain the consistently high temperatures required to form the thin-walled, impermeable vessels (on fuel-types and kilns required for controlled firing above 900°C, see Rye 1981: 96–110). The unique qualities of the raw material permitted a large variety of forms to be produced from the same clay; in this manner, modular techniques could be employed for mass production of pots. Such an approach is diametrically opposed to those used in earlier and contemporary workshops, such as those of EB II Arad, where several ateliers appear to have coexisted, their potters carefully selecting the paste, matching each group of forms (bowls, pots, jars, etc.) to their favored combination of clay and temper (Porat, unpublished; and cf. EB III Numeira and Bab-edh-Dhraᶜ [Schaub 1987]).

The advantages of Metallic Ware for every day use were evidently widely recognized in EB II. Near its presumed production centers, Metallic Ware monopolized ceramic production to such an extent that it is difficult to identify any local industries (with the exception of cooking-pot manufacture) within a radius of some 30 km. The uniformity of the ceramic assemblage at core Metallic Ware sites (such as Tel Dan and Tel Teᵓo) once again stands in contrast to the typically variegated composition of raw materials identified at more southern sites, in the Early Bronze Age as well as in later periods (cf. Wood 1990: 77).

Even at sites located 80–110 km from Mount Hermon, Metallic Ware is dominant, appearing in its full range of forms. The presence of numerous bowls and platters at sites such as Tel Qashish, Qiryat Ata, or Tel Yaqush show that Metallic Ware was distributed, in large quantities, as a commodity in itself, and not merely as a container for other commodities. At greater distances, small quantities of Metallic Ware were distributed either as containers (e.g., "Abydos" jugs) or as fine ware (e.g., a fine platter found at Arad).

Social Context of Metallic Ware Production

In the ongoing discussion concerning the role of pottery analysis in archaeology, we find ourselves firmly allied with those who accord it a primary role, particularly in the study of protohistoric societies. As Knapp (1993: 54) has recently put it, ceramic style and form are "indispensable material proxies for the study of social interaction in complex societies." This ubiquitous component of material culture may be cloned, as it were, to reconstruct a substructure in the fabric of a given ancient society. Its peculiarities of style, raw material, or chronological–geographical distribution may be interpreted to reflect facets of the social context in which it was produced.

The appearance of Metallic Ware coincides with an extraordinary florescence of settlement in all areas where it is abundant. In the Hula Valley alone, recent surveys by I. Shaked and Y. Stepanski have identified more than 20 EB II settlements with Metallic Ware, of which at least 13 were built on previously unoccupied sites—including the urban sites of Tel Dan and Hazor (Greenberg 1990). The northern Golan, where no evidence of EB I settlement has been found, has yielded more than ten EB II sites with Metallic Ware (Hartal 1989). A similar rise has been recorded in surveys of the Lebanese Biqaᶜ and the Galilee highlands (Marfoe 1979; R. Frankel personal communication).

The rapid increase in settlement in northern Canaan has been linked, particularly by Esse (1991: 98–116, 175), to the expansion of trade routes and distribution systems (see also Joffe 1993: 58, 82–83). Leaving aside theoretical issues related to the urbanization process, the general drift of this argument is borne out by the pattern of settlement: almost all sites can be placed near primary or secondary trade routes, and the largest sites are located on the trunk routes, at major transition points between geographical regions (the sites of Bet Yerah, Hazor, Abel Bet Maᶜacah, and Tel Dan, on the easternmost north–south route, may be taken as a case in point).

Concomitantly, the development of craft specialization is often linked to processes of urbanization and expansion of trade. Arnold (1985) comments on the relationship between pottery workshop industries and urbanization, suggesting that the latter acts as a "deviation amplifying mechanism" promoting full-time craft specialization by supplying both an increased demand for pottery and an extensive trade network for its distribution (Arnold 1985: 157, 200). Rice (1984: 47–48) sees ceramic standardization as a means of adapting to rapid population increase.

While one might subscribe to a posited connection between the development of standardized, cost-efficient pottery production and the growth of distribution networks, it can, at best, provide only a partial explanation for the Metallic Ware phenomenon. Urbanization and trade networks characterized all of Canaan during the period in question, in EB III as well as in EB II. Why then did so dominant a ceramic industry emerge only in the far north of the country? What peculiar circumstances permitted its rise in EB II, and its demise in EB III? What made it possible for this industry to extend beyond the usual bounds of the ceramic workshop industry in Bronze Age Canaan? Lastly, where are the antecedents of Metallic Ware, and why do we find it appearing typologically and technologically fully-fledged immediately upon introduction?

The sudden rise in settlement in northern Canaan and adjacent regions, and the appearance of an already established pottery tradition lacking any observable antecedents even in the core areas of Metallic Ware production, suggest a process of colonization. While in southern Canaan, particularly in the middle and lower Jordan Valley, the stages of urbanization can be traced step by step (Schaub 1982; de Miroschedji 1989), this is not the case in the far north. There does not appear to be a broad network of pre-urban settlement, and no site provides us with a sequence of gradual transition from EB I to EB II, such as that observable in the architecture and ceramic assemblages of, for instance, Tell el-Far[c]ah, Jericho, or Bâb edh-Dhra[c]. What seems to be indicated is thus an influx, into the upper Jordan Valley, Upper Galilee, Golan, and Biqa[c], of groups of Canaanite settlers, with well-established architectural (most sites are fortified) and technological (ceramic) traditions.

While the above hypothesis may well explain the uniformity of the Metallic Ware tradition over areas of considerable extent and geographical diversity, it does not yet account for the apparent monopolization of ceramic production by workshops belonging to a single production center, as we suggested. In this regard, if uniform traditions of architecture and ceramic style indicate only a common cultural background, a single production locale for the ceramics of far-flung sites of northern Canaan and adjacent regions suggests an extraordinarily high level of economic integration between them, involving the subordination of local traditions and particular preferences to an overriding, explicit economic interest. And if such a level of integration—indeed, centralization—is indicated, can the commonly accepted, but never explicitly theorized, city-state model for the political and social organization of Canaan accommodate it? Perhaps a model of larger territorial units, with larger-scale economies, should be proposed, if only as a straw man, to test what appear to be implicit assumptions.

To bring out the anomalous character of Metallic Ware production in relation to other industries, a description of its decline may be instructive. At Hazor, where Metallic Ware dominates the earlier phases, the latter part of EB III is represented by several different traditions (Yadin et al. 1961: pls. 154–55): locally produced Khirbet Kerak Ware (of Trans-Caucasian ancestry), an entirely handmade fine ware, polished and fired to temperatures below 700°C (Esse and Hopke 1986); low-fired, thick-walled common local wares, handmade and wheel-finished, characterized by thick peeling slips and pattern burnish; wheelmade wares showing Syrian influence (Yadin et al. 1961: pl. 155:14; Greenberg in press b) and Metallic pithoi (fig. 3:4). Each of these traditions required different techniques for both forming and firing; they also required different kilns and even different fuels. Their presence suggests the existence of competing centers of production, and a demand for elite wares within a context of increasing social differentiation within the towns (Knapp [1993: 73–84] describes such a situation in the city-state system of the central Jordan Valley in MB II). The cosmopolitan character of ceramic production at Hazor, matched at other EB III sites in the north, does not appear to reflect a decline in the fortunes of the city itself. Judging by developments at Bet Yerah, where similar ceramic traditions coexisted within a flourishing EB III metropolis, the opposite may well be true. It does, however, testify to a marked decentralization of ceramic production and a lower level of integration with the neighboring zones, and thus to a significant change in the political and economic organization of the region.

To conclude, the emergence of Metallic Ware, and of many of the sites in which it is abundant, should be seen as a corollary of far-reaching changes affecting Canaan in EB II. Clearly, the powerful forces shaping the picture of urban and rural settlement took different forms of expression in various parts of the land; those characterizing northern Israel and the adjacent region find themselves faithfully, if only partially, portrayed in the character of this ceramic industry, with its centralized production and efficient distribution network. Whether other aspects of production and social relations can be adduced to bolster the preliminary observations suggested here remains to be seen.

ACKNOWLEDGMENTS

We thank the following excavators and researchers for permission to study, analyze, and publish ceramic material and for their cooperation in providing information on contexts and sherd counts: Amnon Ben-Tor, Sharon Zuckerman (Tel Qashish, Tel Yoqneʾam); Emanuel Eisenberg (Tel Teʾo, Bet Yerah); Raphael Frankel (Bet Ha ʿEmeq; Upper Galilee Survey); Ruth Peled and Baruch Brandl, curators of the Israel Antiquities Authority storerooms (Bet Yerah, Rosh Haniqra, Tel Reqet, Amuq wares); Carmela Arnon (Tel Qishyon); Eliot Braun (Meʿona); Amir Golani (Qiryat Ata); Hermann Genz (Kh. ez-Zeraqun); Pirhiya Beck and Lipaz Vinitzky (Lawieh enclosure); Moshe Hartal (Golan Survey). Yuval Goren provided supplemental petrographic data. The Hebrew University, Hebrew Union College, Israel Geological Survey, and the Israel Antiquities Authority provided support for various aspects of our research. We thank friends and colleagues for hours of discussion on the issues presented here. The late Doug Esse provided particular inspiration for many of our ideas, and it is to his memory that this study is dedicated.

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